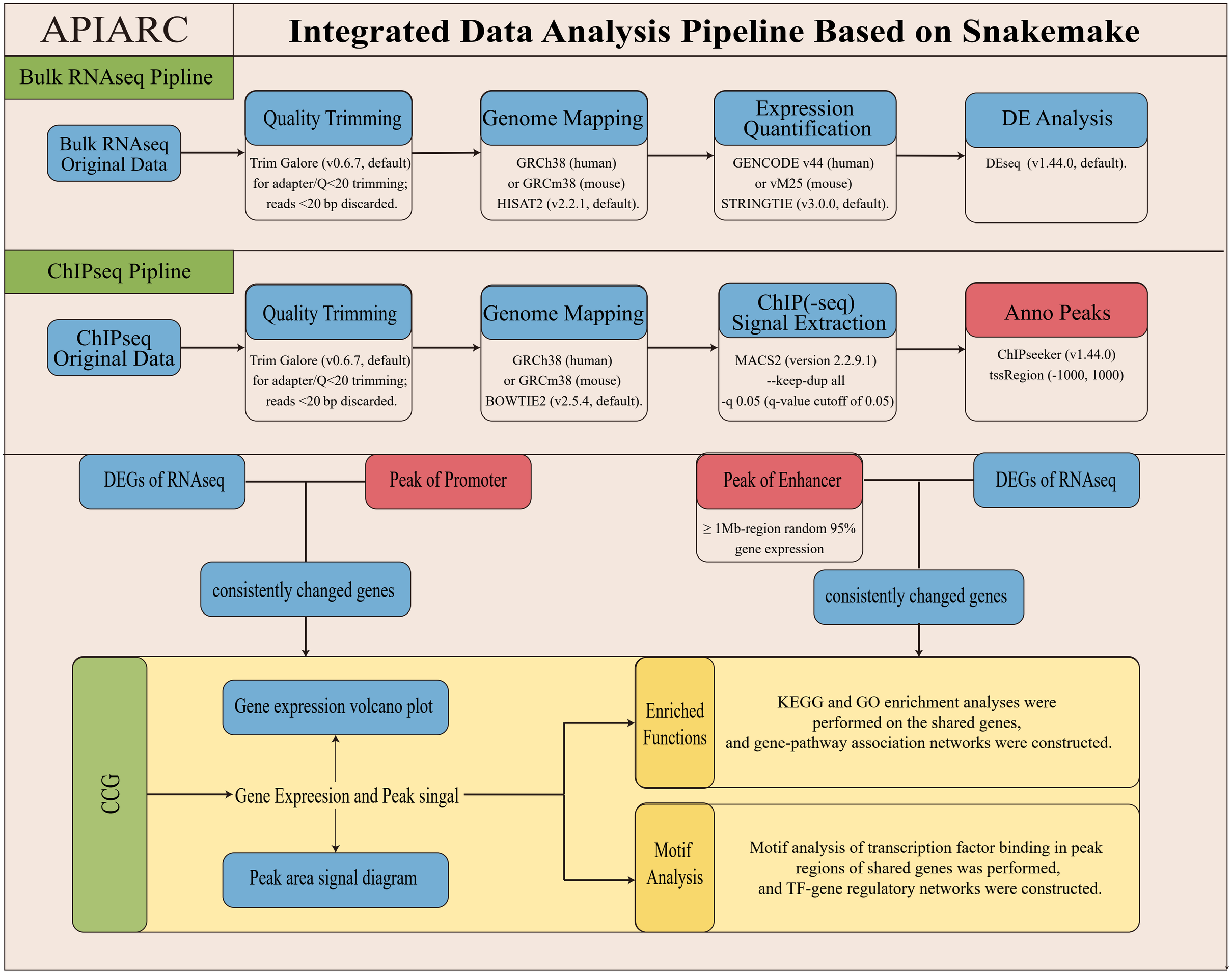
APIARC
Integrated Data Analysis Pipeline Based on Snakemake
Bulk RNAseq Pipline
Bulk RNAseq Original Data
Quality Trimming
Trim Galore (v0.6.7, default) for adapter/Q<20 trimming; reads <20 bp discarded.
Genome Mapping
GRCh38 (human) or GRCm38 (mouse) HISAT2 (v2.2.1, default).
Expression Quantification
GENCODE v44 (human) or vM25 (mouse) STRINGTIE (v3.0.0, default).
DE Analysis
DEseq (v1.44.0, default).
ChIPseq Pipline
ChIPseq Original Data
Quality Trimming
Trim Galore (v0.6.7, default) for adapter/Q<20 trimming; reads <20 bp discarded.
Genome Mapping
GRCh38 (human) or GRCm38 (mouse) BOWTIE2 (v2.5.4, default).
ChIP(-seq) Signal Extraction
MACS2 (version 2.2.9.1) --keep-dup all -q 0.05 (q-value cutoff of 0.05)
Anno Peaks
ChIPseeker (v1.44.0) tssRegion (-1000, 1000)
DEGs of RNAseq
Peak of Promoter
Peak of Enhancer
≥ 1Mb-region random 95% gene expression
DEGs of RNAseq
consistently changed genes
consistently changed genes
CCG
Gene expression volcano plot
Gene Expreesion and Peak singal
Peak area signal diagram
Enriched Functions
KEGG and GO enrichment analyses were performed on the shared genes, and gene-pathway association networks were constructed.
Motif Analysis
Motif analysis of transcription factor binding in peak regions of shared genes was performed, and TF-gene regulatory networks were constructed.